

深圳市金逸晨电子有限公司

LCD MODULE

MODULE NO. :

19264-05 SERIES

Customer:

Approved by:

深圳市金逸晨电子有限公司

Approved by

Checked by

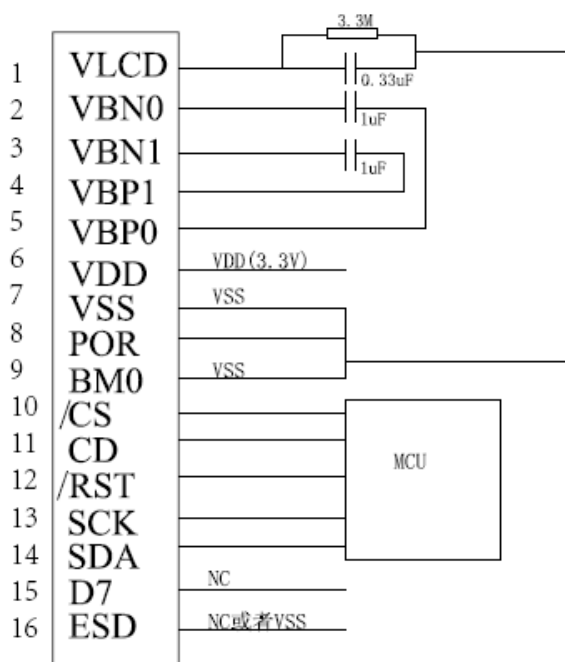
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4. 原理图：



注意：电路及元件值仅作参考

5. 引脚说明：

Pin no.	Symbol	Function
1	VLCD	LCD驱动电压
2~5	VB0-,VB1-, VB1+,VB0+	LCD 偏压
6	VDD	工作电压 $\pm 3.3V$
7	VSS	地
8	POR	上电复位控制
9	BM0	接口选择
10	/CS	片选
11	CD	命令/数据选择
12	/RST	复位
13	SCK	时钟
14	SDA	数据
15	D7	Bi-directional bus. Bit 7.
16	ESD	NC or VSS 空或地
	K	背光源负极(LED-0V)
	A	背光源正极(LED+3V)

Bi-directional bus for both serial and parallel host interfaces.

In serial modes, connect D[0] to SCK, D[5:3] to SDA.

	D7	D6	D5	D4	D3	D2	D1	D0
8-bit (BM=1x)	D7	D6	D5	D4	D3	D2	D1	D0
s8 (BM=00)	--	--	SDA	SDA	SDA	--	--	SCK
S9 (BM=01)	0	--	SDA	SDA	SDA	--	--	SCK
I ² C (BM=01)	1	--	SDA	SDA	SDA	--	--	SCK

For better drive ability, connect D[5:3] together.

Always connect unused pins to either Vss or VDD.

6. 电气特性

6-1 DC电气特性

Item	Symbol	Min	Typ	Max	Unit	Test condition
Operating voltage	V_{DD}	2.6	2.7-3.3	3.6	V	-
Voltage regulator operation voltage	V_0	--	4.8-11.5	11.5	V	-
Dynamic current consumption	I_{DD}	-	135	1000	μA	$V_{DD}=3.0V$, display pattern checker, $T_a=25^{\circ}C$
Sleep mode	I_{DD}	-	0.1	5	μA	$V_{DD}=3.0V$, $T_a=25^{\circ}C$
Standby mode	I_{DD}	-	5	50	μA	$V_{DD}=3.0V$, $T_a=25^{\circ}C$
High-level-Input voltage	V_{IH}	$0.8V_{DD}$	-	V_{DD}	V	A0, D0~D7, WR, CS1, RES,
Low-level Input voltage	V_{IL}	0	-	$0.2V_{DD}$	V	
High-level output voltage	V_{OH}	$0.8V_{DD}$	-	V_{DD}	V	$I_{OH}=-0.5mA$
low-level output voltage	V_{OL}	0	-	$0.2V_{DD}$	V	$I_{OL}=0.5mA$
Input leakage current	I_{IKG}	-1.0	-	1.0	μA	$V_{IN}=V_{SS}$ or V_{DD} (A0, WR, CS1)
LCD driving voltage	V_{LCD}	11.8	12.0	12.2	V	$V_{LCD}=V_O-V_{SS}$

6-2 AC 电气特性

Serial Interface Timing

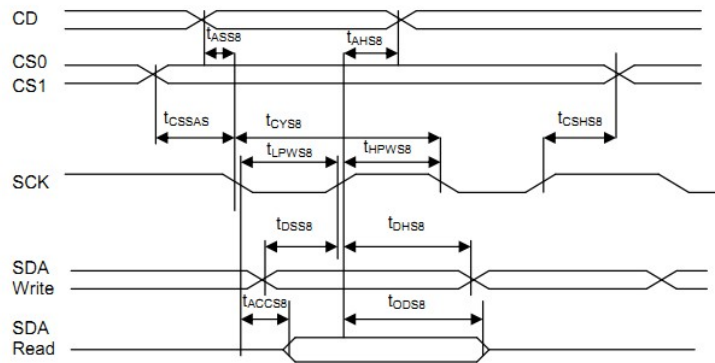


FIGURE 17: Serial Bus Timing Characteristics (for S8)

Symbol	Signal	Description	Condition	Min.	Max.	Unit
(2.5V ≤ V _{DD} ≤ 3.6V, Ta = -30 to +85°C) (Read / Write)						
t _{ASS8}	CD	Address setup time		5	—	nS
t _{AHS8}	CD	Address hold time		10	—	nS
t _{CSSAS8}	CS1, CS0	Chip select setup time		10	—	nS
t _{CSHS8}	CS1, CS0	Chip select hold time		10	—	nS
t _{CVS8}	SCK	System Cycle time		190 / 80	—	nS
t _{LPPWS8}	SCK	Low pulse width		80 / 25	—	nS
t _{HPWS8}	SCK	High pulse width		80 / 25	—	nS
t _{DSS8}	SDA (Write)	Data setup time		25	—	nS
t _{DHS8}	SDA (Write)	Data hold time		10	—	nS
t _{ACCS8}	SDA (Read)	Read access time	C _L = 100pF	—	80	nS
t _{ODS8}	SDA (Read)	Output disable time		—	30	nS

Reset Timing

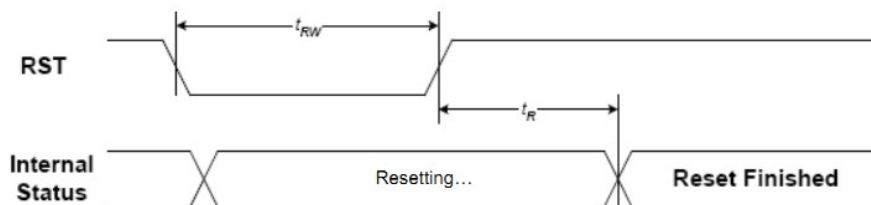


FIGURE 20: Reset Characteristics

(1.7V ≤ V_{DD} ≤ 3.6V, Ta = -30 to +85°C)

Symbol	Signal	Description	Condition	Min.	Max.	Unit
t _{RW}	RST	Reset low pulse width		3	—	μS
t _R	RST, Internal Status	Reset to Internal Status pulse delay		6	—	mS

7. 指令表

The following is a list of host commands supported by UC1609c

C/D: 0 / 1 – Control / Data

W/R: 0 / 1 – Write / Read Cycle

D7~D0: # Useful Data bits, – Don't Care

No	Command	C/D	W/R	D7	D6	D5	D4	D3	D2	D1	D0	Action	Default
1.	Write Data Byte	1	0	#	#	#	#	#	#	#	#	Write 1 byte	N/A
2.	Read Data Byte	1	1	#	#	#	#	#	#	#	#	Read 1 byte	N/A
3.	Get Status	0	1	ID	MX	MY	WA	DE	WS	MD	MS	Get Status PMO[5:0]	N/A
				VER	POR	#	#	#	#	#	#		
4.	Set Column Address LSB	0	0	0	0	0	0	#	#	#	#	Set CA [3:0]	0
	Set Column Address MSB	0	0	0	0	0	1	#	#	#	#	Set CA [7:4]	0
5.	Set Temp. Compensation	0	0	0	0	1	0	0	1	#	#	Set TC[1:0]	00b
6.	Set Power Control	0	0	0	0	1	0	1	#	#	#	Set PC[2:0]	110b
7.	Set Adv. Program Control (double-byte command)	0	0	0	0	1	1	0	0	R	R	Set APC[R][7:0], R = 0~3	N/A
				#	#	#	#	#	#	#	#		
8.	Set Scroll Line	0	0	0	1	#	#	#	#	#	#	Set SL[5:0]	0
9.	Set Page Address	0	0	1	0	1	1	#	#	#	#	Set PA[3:0]	0
10.	Set V _{BIAS} Potentiometer (double-byte command)	0	0	1	0	0	0	0	0	0	1	Set PM[7:0]	49H
				#	#	#	#	#	#	#	#		
11.	Set Partial Display Control	0	0	1	0	0	0	0	1	0	#	Set LC[5]	0b
12.	Set RAM Address Control	0	0	1	0	0	0	1	#	#	#	Set AC[2:0]	001b
13.	Set Frame Rate	0	0	1	0	1	0	0	0	#	#	Set LC[4:3]	01b
14.	Set All-Pixel-ON	0	0	1	0	1	0	0	1	0	#	Set DC[1]	0b
15.	Set Inverse Display	0	0	1	0	1	0	0	1	1	#	Set DC[0]	0b
16.	Set Display Enable	0	0	1	0	1	0	1	1	1	#	Set DC[2]	0b
17.	Set LCD Mapping Control	0	0	1	1	0	0	0	#	#	0	Set LC[2:1]	00b
18.	System Reset	0	0	1	1	1	0	0	0	1	0	System Reset	N/A
19.	NOP	0	0	1	1	1	0	0	0	1	1	No operation	N/A
20.	Set Test Control (double-byte command)	0	0	1	1	1	0	0	1	TT		For testing only. Do not use.	N/A
				#	#	#	#	#	#	#	#		
21.	Set LCD Bias Ratio	0	0	1	1	1	0	1	0	#	#	Set BR[1:0]	11b: 9
22.	Set COM End	0	0	1	1	1	1	0	0	0	1	Set CEN[5:0]	63D
				--	--	#	#	#	#	#	#		
23.	Set Partial Display Start	0	0	1	1	1	1	0	0	1	0	Set DST[5:0]	0
				--	--	#	#	#	#	#	#		
24.	Set Partial Display End	0	0	1	1	1	1	0	0	1	1	Set DEN[5:0]	63D
				--	--	#	#	#	#	#	#		
25.	Set MTP Operation Control	0	0	1	1	1	1	1	0	0	0	Set MTPC[4:0]	00H
				--	--	--	#	#	#	#	#		
26.	Set MTP Write Mask	0	0	1	1	1	1	1	0	0	1	Set MTPM[5:0]	0
				--	--	#	#	#	#	#	#		
27.	Set V _{MTP1} Potentiometer	0	0	1	1	1	1	0	1	0	0	Set VMTP1[7:0]	N/A
				#	#	#	#	#	#	#	#		
28.	Set V _{MTP2} Potentiometer	0	0	1	1	1	1	0	1	0	1	Set VMTP2[7:0]	N/A
				#	#	#	#	#	#	#	#		
29.	Set MTP Write Timer	0	0	1	1	1	1	0	1	1	0	Set MTPWT[7:0]	N/A
				#	#	#	#	#	#	#	#		
30.	Set MTP Read Timer	0	0	1	1	1	1	0	1	1	1	Set MTPRT[7:0]	N/A
				#	#	#	#	#	#	#	#		

Serial Read Command (Enabled only in S8/S9 mode)

31.	Get Status	0	0	1	1	1	1	1	1	1	0	Get Status PMO[5:0]	N/A
		0	1	ID	MX	MY	WA	DE	WS	MD	MS		
				VER	POR	#	#	#	#	#	#		
32.	Read Data	0	0	1	1	1	1	1	1	1	1		FFH
		1	1	#	#	#	#	#	#	#	#		

Any bit pattern other than those listed above may result in NOP (No Operation).